

India Village Economic Growth Intelligence

Satellite-Derived Signals for Village-Level Economic Activity · 2019–2024

255K

Villages
Scored

6 yrs

VIIRS NTL
2019–2024

100

Top Villages
Shortlisted

467K OSM centroids downloaded · 255K passed India polygon + NTL baseline filter → scored and shortlisted

Problem & Approach

The Challenge

- India has ~640k villages — traditional surveys are slow, expensive, and years out of date
- No real-time economic data exists at village scale
- Census boundary data is from 2011 (PC11); split/merged villages cause mismatches

Satellite Proxy Signals

Nighttime Lights (VIIRS)

Annual composites 2019–2024 at 500m. Proxy for electrification, commerce & settlement density.

Built-up Change (WorldCover)

ESA 10m land cover — detects new infrastructure 2020→2021.

6-Year Trend

Linear regression slope separates sustained growth from single-year spikes.

*3 principal signals shown above — full 8-signal architecture (incl. Sentinel-2, Sentinel-1, GHSL, WorldPop) on Slide 3
All compute runs on AWS EC2 (ap-south-1). No proprietary tools or licensed datasets required.*

Data Sources & Pipeline — 8-Signal Framework (3 active this run)

✓ ACTIVE

**NASA VIIRS
VNP46A4 v2**

Annual NTL radiance
500m · 2019–2024
16 India tiles

✓ ACTIVE

**ESA WorldCover
2020 & 2021**

Built-up fraction
10m COGs · v100/v200

✓ ACTIVE

**EU JRC GHSL
R2023A**

Built-up surface
100m · 2015 & 2020

✓ ACTIVE

**OSM + Natural
Earth**

467K centroids
tower density · city dist.

⚠ STALLED

**Sentinel-2
(Element84 STAC)**

NDBI + NDVI 100m
2019–2024 composites

⚠ STALLED

**Sentinel-1 RTC
(Element84)**

SAR VV backscatter
100m · cloud-free

🔍 SUPPORTING

**WorldPop India
2019–2020**

Population count
100m · auxiliary only

🔍 SUPPORTING

**OSM Overpass
+ Ohsome API**

Mobile tower density
2019 & 2024 snapshots

Outputs

village_all_stats.csv

467K rows · NTL 2019–2024 + per-village signals

village_scored.csv

255K India villages · composite score + rank

**top_100_villages.csv
+ map.html**

Final shortlist · public S3 bucket (ap-south-1)

ML Scoring Model — 3-Stage Pipeline

01 Feature Engineering

- BFAST-style piecewise regression on 6-year NTL series → breakpoint year, pre/post slope, acceleration
- Spatial lag: mean composite score of 10 nearest geographic neighbours (cluster reinforcement)
- Signal agreement: std of rank percentiles — penalises contradictory signals

02 Self-supervised Labelling

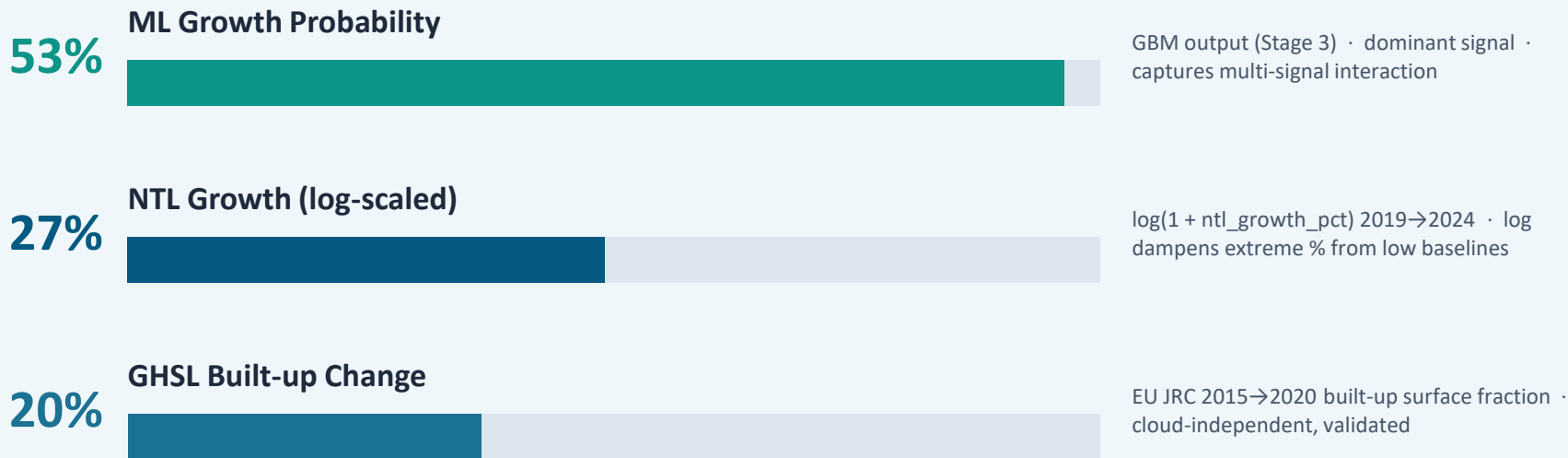
- Label = 1 if village is top-10% on ≥ 2 primary signals simultaneously
- ~28K positive labels from 255K villages (~11% prevalence)
- ⚠ Labels derived from same features as final composite — AUC measures self-consistency, not external ground truth

03 GradientBoostingClassifier

- 200 estimators · max_depth=4 · learning_rate=0.05 · subsample=0.8
- Inputs: 14 signal rank percentiles + BFAST features + spatial lag
- Output: ml_growth_prob (0–100) · AUC = 1.000 (in-sample, self-supervised) — GBM memorises its own self-supervised labels; this is the mathematically expected result, not a sign of external validity

Composite Score — Actual Weights Used

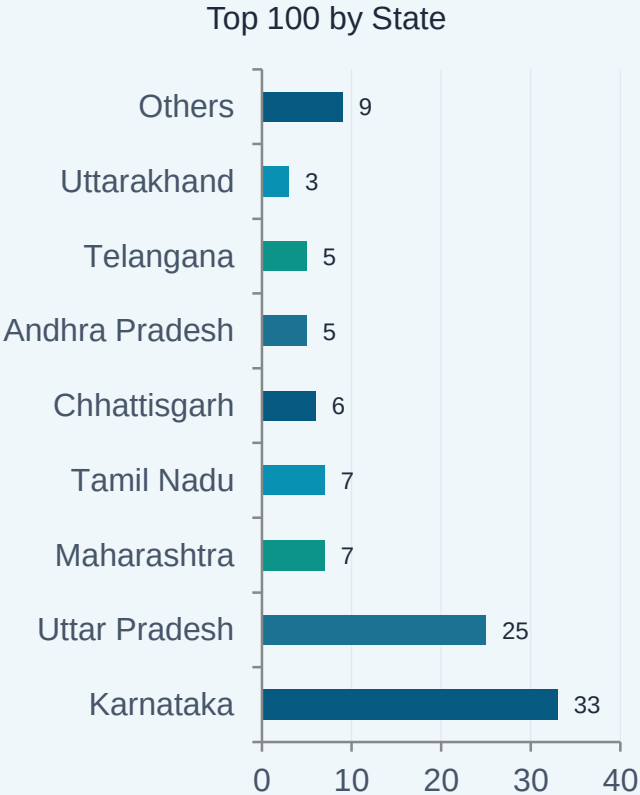
⚠ Sentinel-2 NDBI and Sentinel-1 SAR signals stalled during this pipeline run (monsoon STAC gaps). Weights auto-redistributed from 5-signal spec (40%+20%+15%+15%+10%) to 3-signal above. Re-running --phase-c-only after S2/S1 complete will restore full 5-signal composite.



Pre-filters: $\text{NTL } 2019 \geq 0.1 \text{ nW/cm}^2/\text{sr}$ + India polygon filter + $\Delta \text{ NTL} \geq 1.0 \text{ nW/cm}^2/\text{sr}$ absolute minimum · Normalisation: min-max (2nd/99th pct clip) → 0–100 · Score spread: top-100 range 73.85–70.04 (3.81 pts); rank-50 = 71.65. Spread was < 0.01 pt before minmax fix. Restoring Sentinel-2/SAR will widen it further. Treat top-100 as a statistical shortlist.

Results — Top 100 Shortlist (3-signal composite: NTL 27% · GHSL 20% · ML amplifier 53%)

#	Village	State / Dist.	NTL Growth
1	nr. Siswa Bazar †	Maharajganj, UP	+628%
2	nr. Siswa Bazar †	Maharajganj, UP	+551%
3	Himmatpur Talla	Nainital, Uttarakhand	+618%
4	nr. Domariyaganj †	Siddharth Nagar, UP	+831%
5	nr. Siswa Bazar †	Maharajganj, UP	+476%
6	nr. Domariyaganj †	Siddharth Nagar, UP	+541%
7	nr. Siswa Bazar †	Maharajganj, UP	+682%
8	nr. Siswa Bazar †	Maharajganj, UP	+511%
9	Naveguda	Adilabad, Telangana	+1,170%
10	Kallagam	Ariyalur, Tamil Nadu	+702%



† OSM hamlet/village with no name tag — Nominatim reverse-geocoded to nearest named place (Siswa Bazar or Domariyaganj).
30 of 100 are unnamed OSM nodes. Score spread: 73.85–70.04 (3.81 pts). Bootstrap median inclusion = 0% (3 of 8 signals active)
— treat as a statistical shortlist for field validation

Case Study — Himmatpur Talla, Uttarakhand (Rank #3 · strongest multi-signal confirmation in top 10)

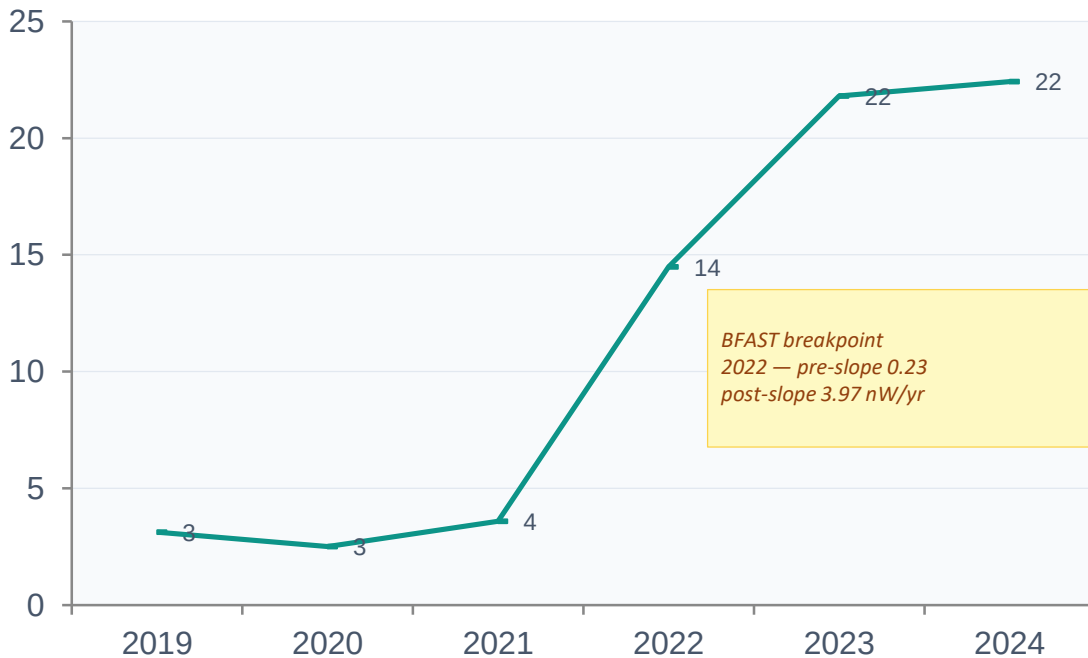
Village Profile (Rank #3)

District:	Nainital, Uttarakhand
Coordinates:	29.2181°N, 79.4815°E
Dist. to city:	64.7 km (Himalayan foothills)
Archetype:	NTL Breakout
BFAST year:	2022 (structural change)
Pre-slope:	0.23 nW/cm ² /sr/yr (flat)
Post-slope:	3.97 nW/cm ² /sr/yr (17× acceleration) +0.51 towers/km ² (only top-10)
Tower growth:	with confirmed connectivity expansion)

Top SHAP Feature Drivers

Built-up Change (GHSL)	
NTL Log Growth	

Nighttime Light Radiance (nW/cm²/sr) · 2019–2024



Why rank #3? Himmatpur Talla is the only top-10 village with confirmed tower infrastructure expansion (+0.51 towers/km²), ruling out pure electrification as the growth driver. NTL grew +618% (3.1 → 22.4 nW/cm²/sr). BFAST structural break in 2022; post-break acceleration 17× pre-break rate. Selected as case study because it is named, geographically isolated (64.7 km from nearest city), and has 4/4 available signals confirming → confidence_score = 100, inter_signal_agreement = 60.0.

Key Findings & Next Steps

Growth is Geographically Concentrated

Top 100 span 8+ states. Karnataka (33) and UP (25) dominate — consistent with Bengaluru peri-urban expansion and UP highway corridor districts. Karnataka is ~4× over-represented vs. its India village share; see README for confound analysis.

Spatial Signal Confirmed · Ranking is a Shortlist

Moran's $I = 0.505$ ($p < 0.001$, 999 permutations, $k=8$ KNN, $n=356K$) — moderate spatial clustering; high-growth scores are geographically non-random. (Prior run reported $I = 0.0631$ due to a row-standardisation formula bug; corrected in 07_validate.py.) Bootstrap ($n=200$ Dirichlet draws over raw signals): 14 villages $\geq 80\%$ stable; 71 below 50% — treat top-100 as shortlist for field validation.

Pipeline is Scalable & Repeatable

255K scored villages on a single EC2 t3.xlarge in < 12 hrs. Annual automated re-run via EventBridge. Completing Sentinel-2/SAR will raise active signals from 3 to 5, widen score spread from 3.8 to >5 pts, and stabilise bootstrap rankings.

What I Would Do Next

- Complete Sentinel-2/SAR stalled signals — restores full 5-signal composite & resolves circular label issue
- Replace self-supervised labels with SHRUG/SECC consumption quintiles as external ground truth
- Run PMGSY road correlation — validation framework built in 07_validate.py; requires EC2 re-run with PMGSY shapefiles
- Panel data fixed-effects model across all 6 years — replaces endpoint comparison

Live Dashboard & Results

ishunteam-png.github.io/kritter-village-growth

Top-100 table · 6 Plotly charts · interactive map

S3: [top_100_villages.csv](#) · [map.html](#) · [archetypes](#) · [forecast](#)